

C++ Programming Course Content 1. Introduction to C++ - History and features of C++ - Installing compiler and IDE (GCC, Visual Studio, CodeBlocks) - Structure of a C++ program - Compilation and execution process 2. Basic Concepts - Variables and data types - Constants and literals - Operators (Arithmetic, Relational, Logical, Bitwise, Assignment) - Input/Output (cin, cout) 3. Control Statements - if, if-else, nested if - switch-case - Loops: for, while, do-while - break, continue, goto 4. Functions - Function declaration and definition - Call by value and call by reference - Inline functions - Recursion 5. Arrays and Strings - One-dimensional and multi-dimensional arrays - String handling using char arrays - String class in C++ 6. Pointers - Pointer basics - Pointer arithmetic - Pointers and arrays - Dynamic memory allocation (new, delete) 7. Object-Oriented Programming (OOP) - Classes and objects - Constructors and destructors - Encapsulation and data hiding - Access specifiers (public, private, protected) 8. Inheritance - Single, multiple, multilevel, hierarchical inheritance - Method overriding - Virtual functions and polymorphism 9. Polymorphism - Function overloading - Operator overloading - Runtime polymorphism 10. Templates and STL - Function templates - Class templates - Standard Template Library (STL): vector, list, map, set, stack, queue 11. Exception Handling - try, catch, throw - Multiple catch blocks - User-defined exceptions 12. File Handling - Reading and writing files - fstream, ifstream, ofstream 13. Advanced Topics - Namespaces - Lambda expressions - Smart pointers - Multithreading basics 14. Mini Projects - Student Management System - Bank Management System - Library Management System

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante. Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc,